

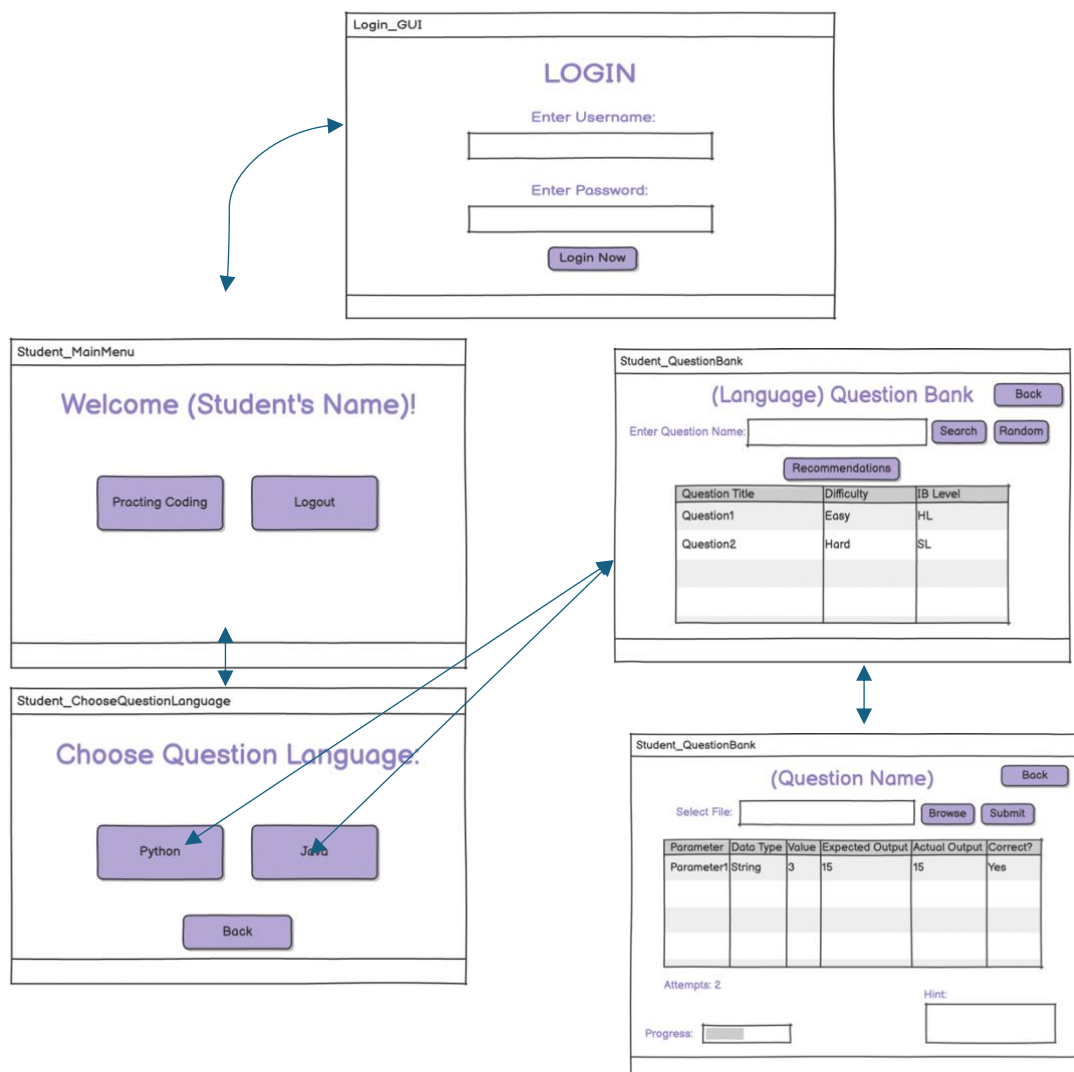
Computer Science Internal

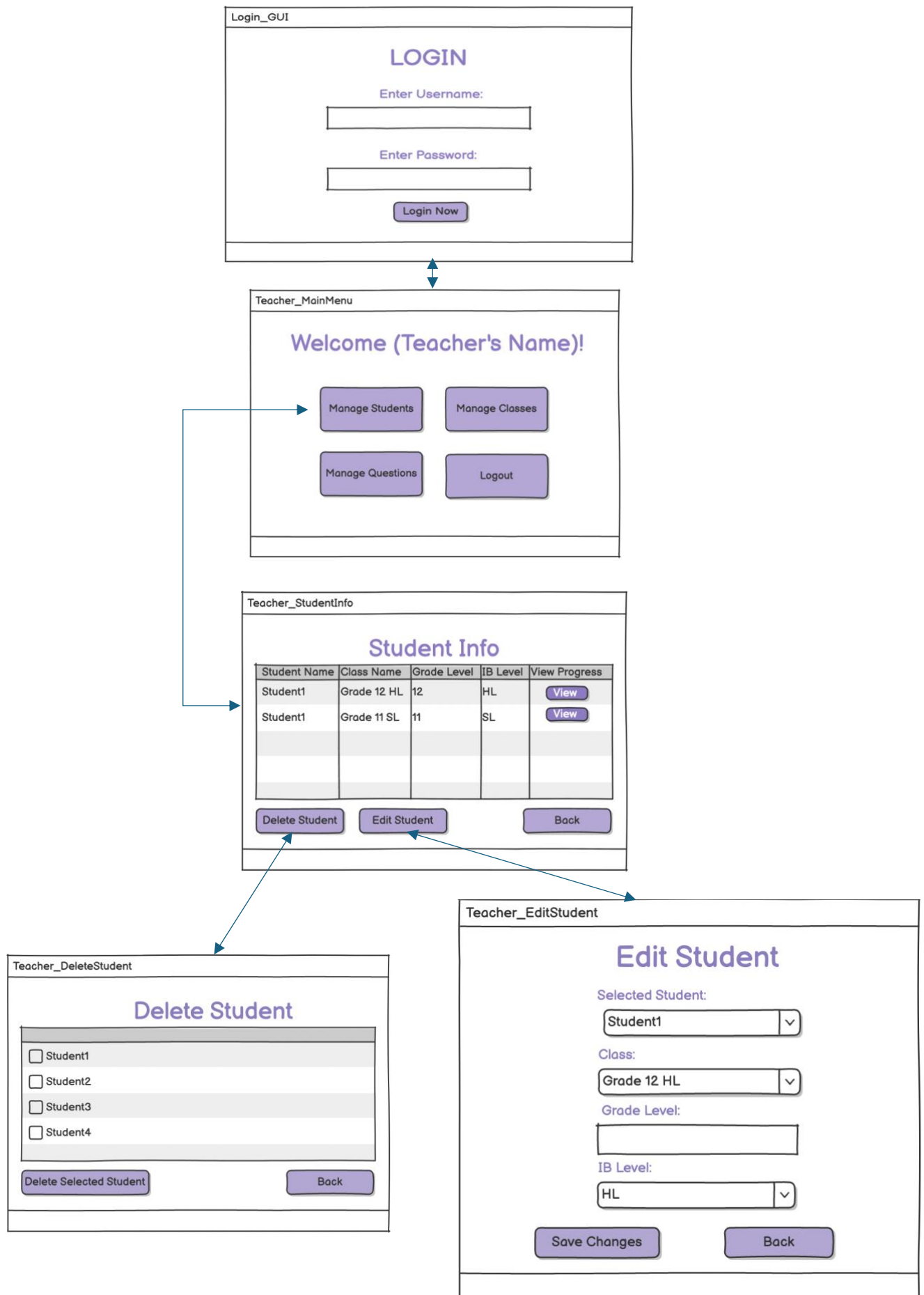
Assessment: Criterion B – Design

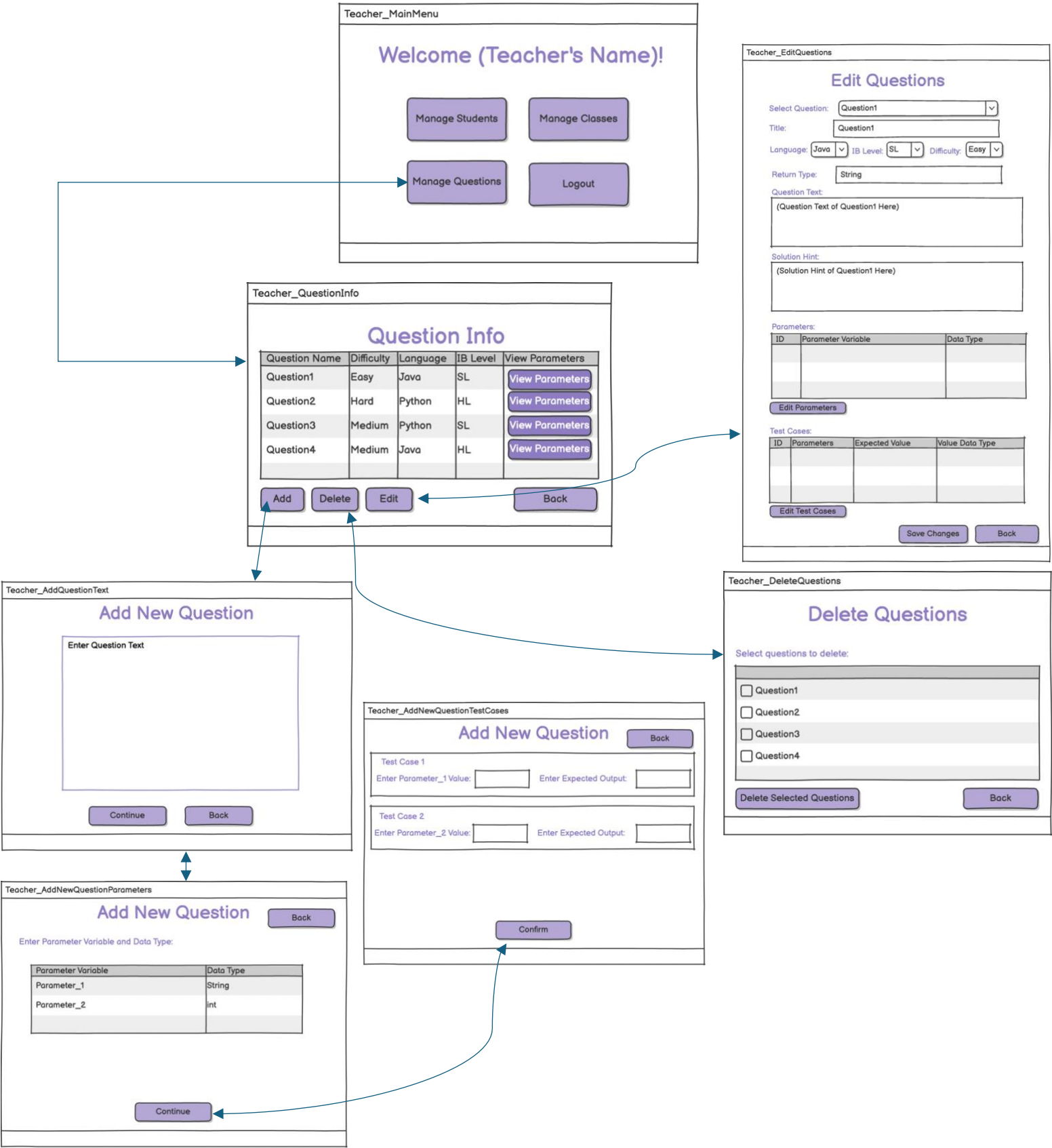
Total Word Count: 48

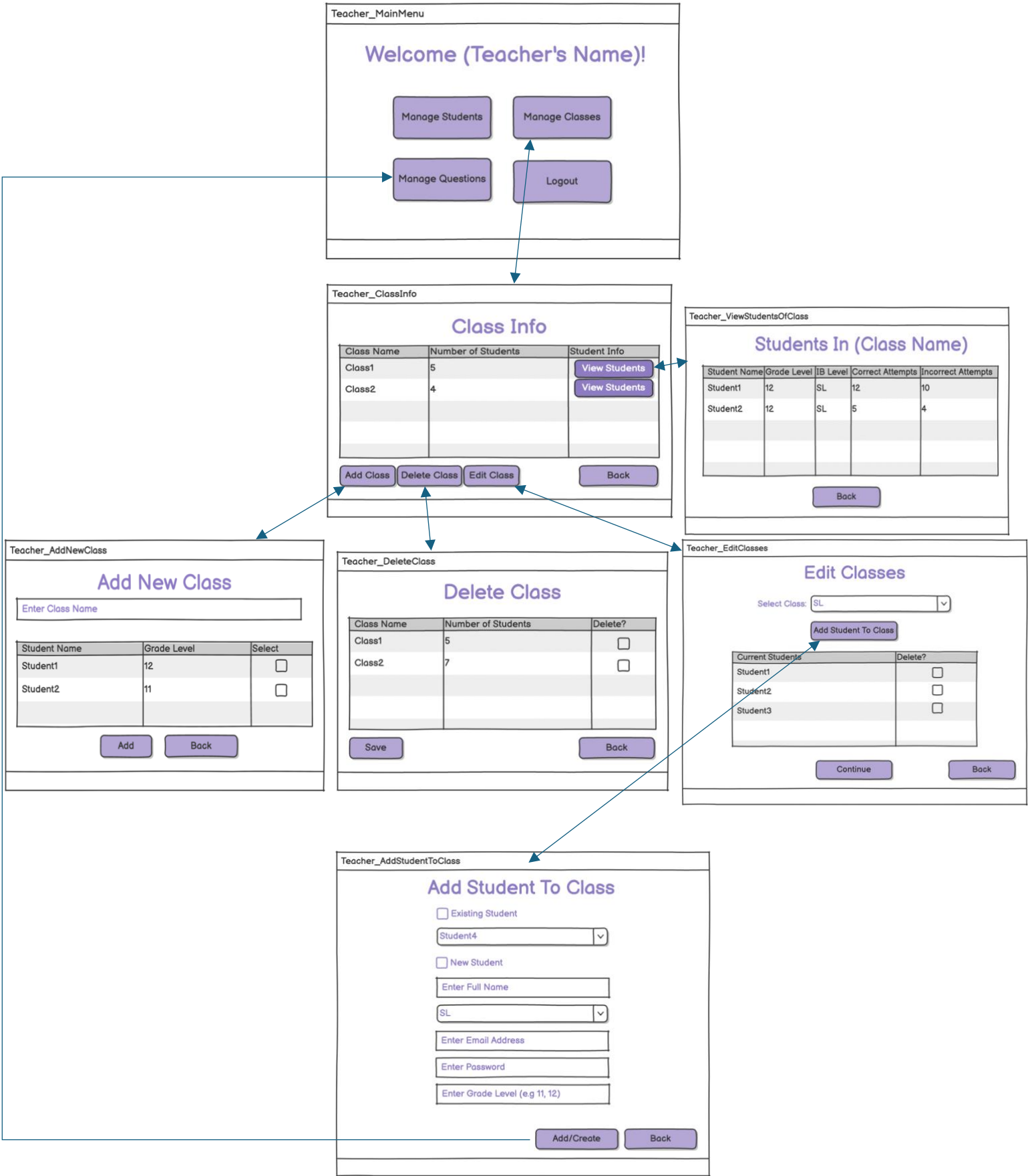
GUI Design:

I used Balsamiq to create a detailed design prototype. The client wanted to change the “Back” buttons to red and “Continue” buttons to green (Appendix B-3), and this was adjusted for the final design.



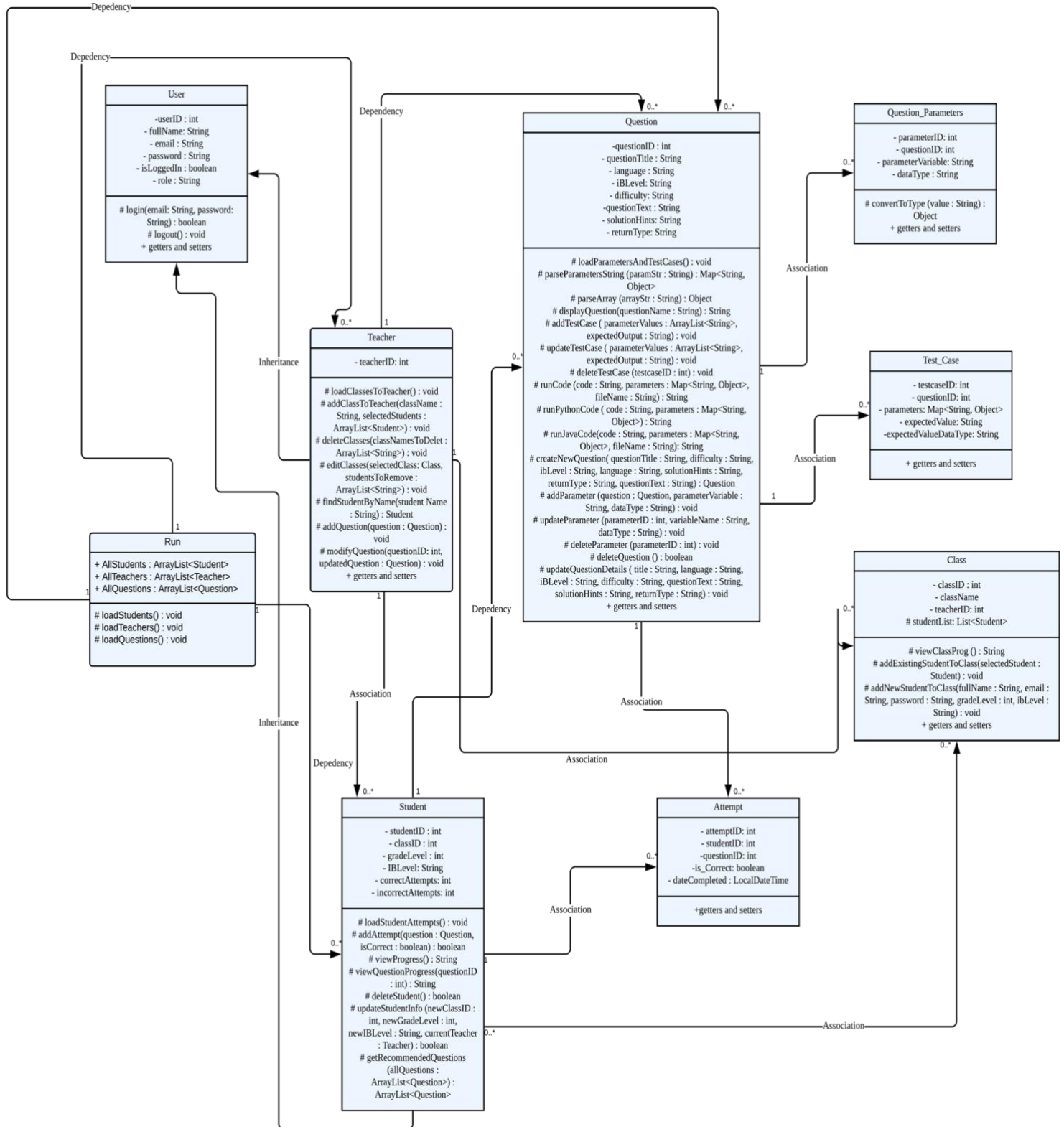




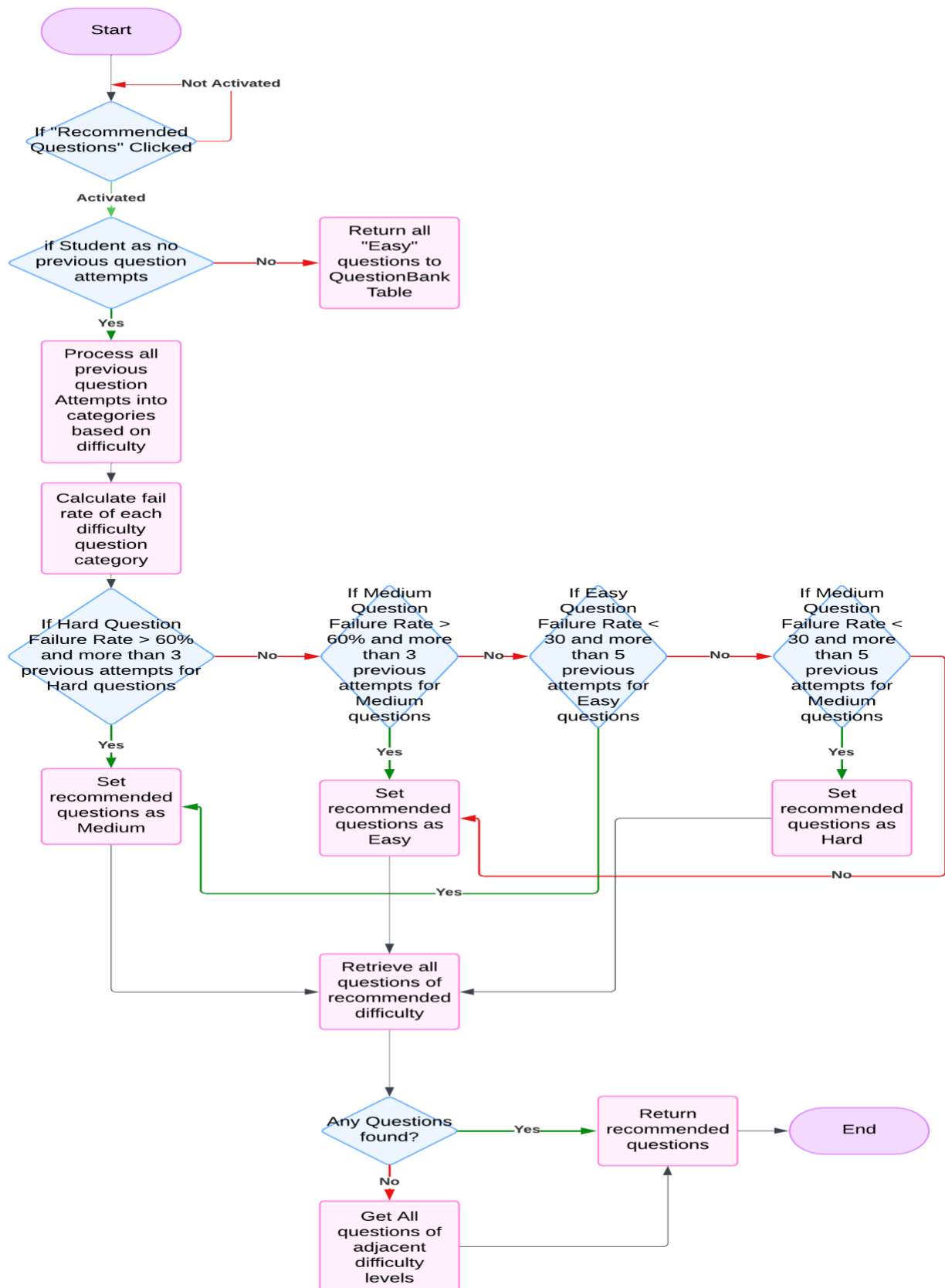


UML Diagram and Flow Charts:

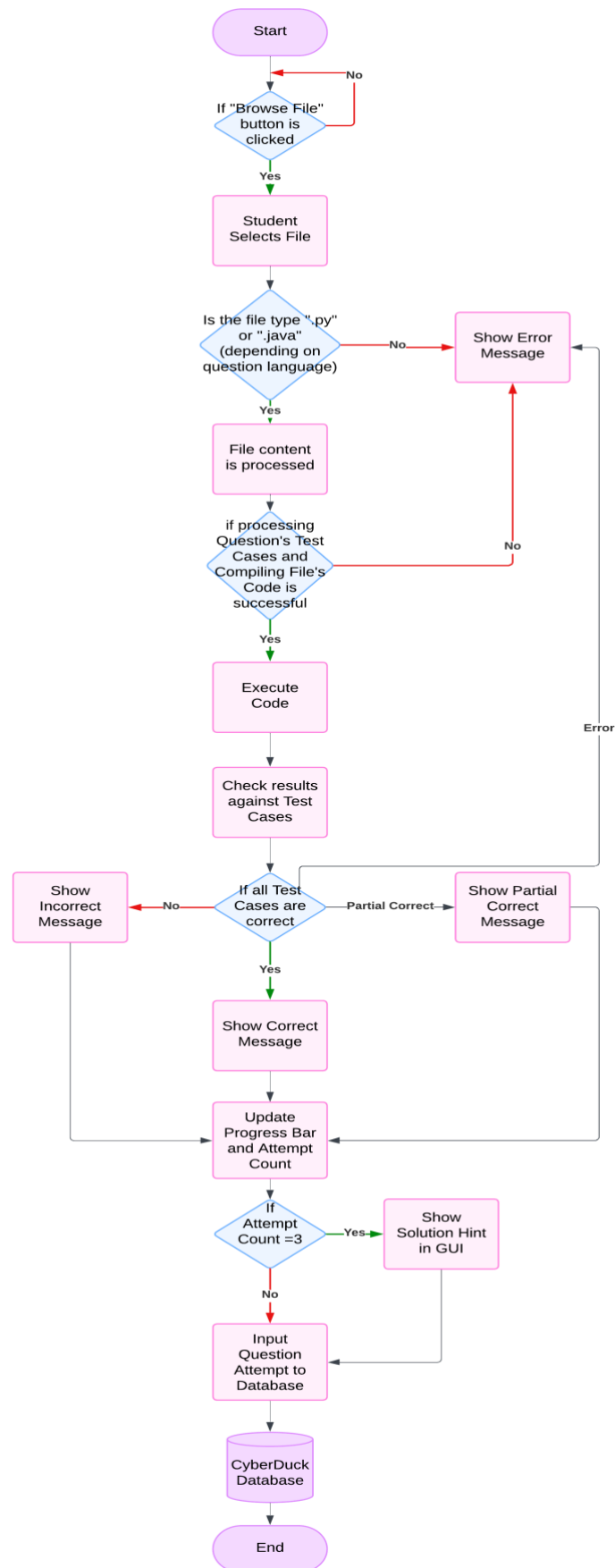
Using Lucid, I created the UML diagram and the flow charts for the program



Recommended Questions Algorithm:



Submit Answer and Running Code:



Database Structures:

Excel Draft Structure:

Classes Table		
classID	teacherID	className
301	101	Grade 11 SL
302	102	Grade 12 HL
303	103	Grade 12 SL

Users Table					
userID	fullName	email	password	role	
1	John Doe	john.doe@gmail.com	\$argon2i\$v=19\$m=65536,t=10,p=1\$QhNbZEbnR9s8n/aS65W7IA\$Cp/lfdOKB5GyZYDf1a0hPls5hH/INa8Vhctw/so4RB0	teacher	
2	Jane Smith	jane.smith@gmail.com	\$argon2i\$v=19\$m=65536,t=10,p=1\$GEOETV9/L0B9vhFm2XpoZA\$dovuXOPSIoCCNWKPS4gZncU+Q5wFqXN/CIM9gNy	student	
3	Scott Brown	sb@gmail.com	\$argon2i\$v=19\$m=65536,t=10,p=1\$QhNbZEbnR9s8n/aS65W7IA\$Cp/lfdOKB5GyZYDf1a0hPls5hH/INa8Vhctw/so4RB0	teacher	
4	Bob Johnson	bob.johnson@gmail.com	\$argon2i\$v=19\$m=65536,t=10,p=1\$5vFH+rHJgZcyjPovtAFU+Q\$FZGxdZKa89jLcXaZleiv/gdD2MVRcWXAj7KvWjXNio0	teacher	
5	Gabriella Aguire	gabi.a@gmail.com	\$argon2i\$v=19\$m=65536,t=10,p=1\$cWKQpP7XCciudknTEp+Kw\$4P+ivL2igDtqPvRk0wSoUgOumlKGU/l+db9MTMOgm5	student	

Teachers Table		
userID	teacherID	
1	101	
3	102	
4	103	

Question Parameters Table			
parameter	questionID	parameterVariable	dataType
501	401	a	int
502	401	b	int
503	402	n	long
504	403	str	String

Students Table						
studentID	classID	gradeLevel	IBLevel	correctAttempts	incorrectAttempts	userID
201	NULL		11 SL	10		2
202	302		12 HL	13	15	5

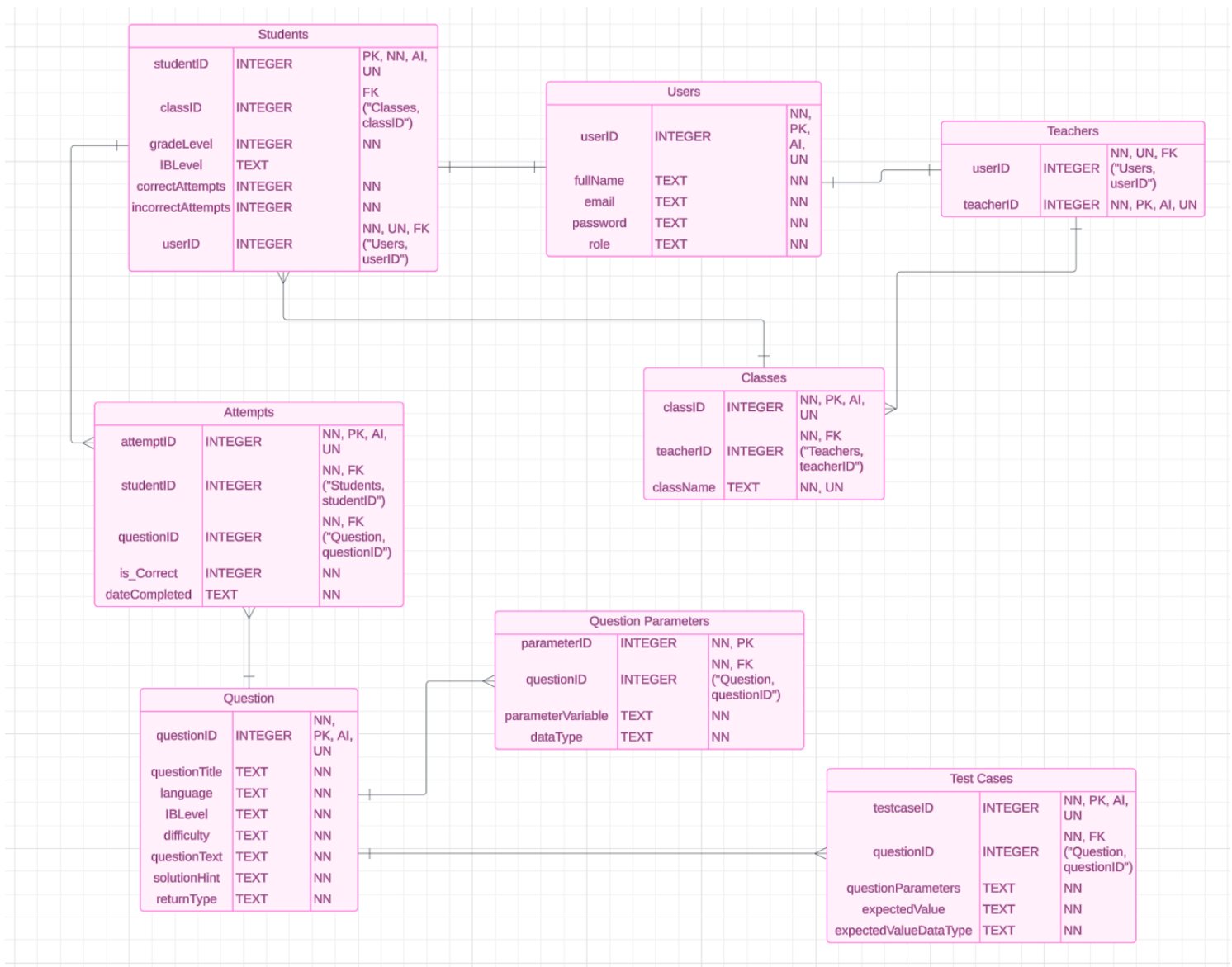
Test Cases Table				
testcaseID	questionID	questionParameters	expectedVal	expectedValueDataType
601	402	{"n": 5}	120	long
602	401	{"a": 5, "b": 3}	8	int
603	401	{"a": -1, "b": 1}	0	int
604	403	{"str": "Hello"}	HHeellllloo	String
605	403	{"str": "Hi-there"}	HHii--tthheerre	String

Attempts Table				
attemptID	studentID	questionID	is_Correct	dateCompleted
701	201	401	1	2024-09-24T06:45:30
702	201	402	1	2024-09-24T07:15:22
703	202	401	1	2024-09-24T08:30:00
704	202	403	0	2024-09-24T09:05:45

Question Table				
questionID	questionTitle	language	IBLevel	difficulty
401	Sum_Of_Two_Numbers	Java	SL	Easy
402	Factorial_Calculation	Python	HL	Medium
403	double_Char	Java	SL	Hard

questionText	solutionHint	returnType
Write a function to calculate the sum of two numbers.	Use the '+' operator	int
Calculate the factorial of a number.	Use recursion or iteration!!	long
Given a string, return a new string where each character in the original appears twice in a row	Think about building a new string by processing each character one at a time	String

Database Structure Relationships:



Test Plan:

Test Number	Action to Test	Method of Testing	Expected Outcome	Success Criteria	Tested	Fulfilled
1	Login – Invalid Credentials + Validation Rules	Enter various combinations of incorrect username/email and/or password	System denies access and allows incorrect credentials to be handled	1.a + 1.c		
2	Login – Valid Student	Enter correct Student credentials	System verifies credential and provides access to Student Main Menu correlated to the Student credentials	1.b + 1.d		
3	Login – Valid Teacher	Enter correct Teacher credentials	System verifies credential and provides access to Teacher Main Menu correlated to the Teacher credentials	1.b + 1.e		
4	Student – Choose Question Language	Navigate to StudentQuestionBank GUI by clicking “Practice Coding” button	System provides both Java and Python QuestionBank GUIs and presents their associated language questions	2.a.i		
5	Recommended Questions Algorithm	Complete several questions (both correctly and incorrectly)	Verify the recommendation algorithm adapts to Student’s attempts in questions, providing different recommended questions when checked at different times	2.a.iii		
6	Student – Testing Student Code	Submit a different samples of student question answers (some correct, some partially correct, and some completely wrong)	The test case table’s results align with the respective submitted code	2.a.ii		
7	Progress Bar during Question	Submit partially correct code	The progress bars aligns with the number of successful test cases	2.a.iv		

8	Solution Hint Box	Make 3 failed attempts in a question	The hint appears only after the 3 failed attempts	2.a.v		
9	Teacher – Manage Questions	Click on the Manage Questions button and navigate through the Teacher interface to add, edit, and delete questions in the question's dashboard	The added, edited, and deleted questions should be reflected within the CyberDuck database and in the questions table.	2.b.i		
10	Teacher – Manage Students	Click on the Manage Students button and navigate through the Teacher interface to edit and delete students in the students dashboard and click the “View Progress” button to view individual student progress	The edited and deleted questions should be reflected within the CyberDuck database and the students table, and individual student progress is shown after clicking the “View Progress” button by displaying their individual attempts of questions and number of correct and incorrect attempts	2.b.ii		
11	Teacher – Manage Classes	Click on the Manage Classes button and navigate through the Teacher interface to add, edit, and delete classes in the classes' dashboard	The added, edited, and deleted classes should be reflected within the CyberDuck database and in the classes table.	2.b.iii		